Coherent Olympic Forecasts through National Opportunity Portfolios

The Olympic medal table is a competition for a changing set of opportunities. We model gold and non-gold awards separately within sports, then combine them into coherent country totals. Historical team medals are deduplicated, country totals are reconciled to the supplied official table, and recent sport capacity is partially pooled toward prior athlete exposure.

Our conditional 2028 projection places the United States first with **45.6 gold and 139.9 total medals**; the respective 90% prediction intervals are [32, 60] and [115, 166]. China follows with 36.7 gold and 88.1 total medals. France is projected to fall from 64 total medals in 2024 to 41.8 as its host advantage expires. The strongest predicted improvement is the United States; its probability of exceeding its 2024 total is 80.7%.

The programme file actually ends in 2024, despite its description. These forecasts therefore condition on the 2024 event programme and NOC roster, with the United States hosting. We give the full country table with intervals and separate event-change scenarios. One extra event helps countries in proportion to their sport-specific capacity, not their overall medal-table rank.

Among 65 participating nations with no previous medal in the supplied official history, the model expects **5.38 first-time medallists**, with a 90% prediction interval of 2–9. This forecast is particularly sensitive to the small-country prior and deserves less confidence than the leading-country ordering.

Numerical calibration used targets before 2020. Reused chronological evaluations cover 2020 and 2024; their feedback had been seen during later model development. Total-medal mean absolute errors were 2.09 and 1.48, versus 2.19 and 1.77 for persistence. We screened sustained country-sport performance jumps compatible with coaching improvements, but coach identities are absent and causal effects cannot be isolated. A stated 25% rise in sport odds adds 0.50, 0.41, and 0.81 expected medals for India shooting, Mexico Aquatics (a diving proxy), and Brazil gymnastics. These are investment scenarios, not estimated causal gains.

Keywords: Olympic medals; event portfolio; partial pooling; prediction intervals; first medal; coach attribution.

Contents

1 The decision problem and what the data can support . . . 3
1.1 Data audit and identity policy . . . 3
1.2 Working assumptions . . . 4
2 A coherent model of country-sport medal opportunities . . . 4
2.1 Why allocate opportunities rather than regress ranks? . . . 4
2.2 Recency, exposure, and hosting . . . 5
2.3 Prediction intervals and their interpretation . . . 6
2.4 Parameter selection and chronological evaluation . . . 6
3 Conditional medal table for Los Angeles 2028 . . . 8
3.1 Leading countries and expected changes . . . 8
3.2 The next first-medal nations . . . 9
4 How the programme changes the medal table . . . 10
4.1 A country's important sports are a portfolio . . . 10
4.2 Event number, event type, and a host's choices . . . 12
5 What can be learned about a great-coach effect? . . . 12
5.1 A change screen with a clear identification limit . . . 12
5.2 Three investment proposals and their conditional impact . . . 14
6 Original committee insights: protect opportunities, not just ranks . . . 15
7 Verification, sensitivity, and limits . . . 16
7.1 Independent numerical evidence . . . 16
7.2 Structural sensitivity and decisions that can change . . . 16
8 Conclusion . . . 17
A Complete conditional 2028 forecast . . . 17
Report on Use of AI Tools . . . 21

1 The decision problem and what the data can support

The task is to forecast at least gold and total medals, include countries with no previous medals, explain the role of event composition, examine a possible great-coach effect, and extract actionable insights for Olympic committees [1]. We treat a medal as one country award in an event, so a football team does not receive eleven country medals. All numerical inputs come from the supplied files [2]. The authoritative targets are country gold and total counts; athlete records supply the distribution across sports and evidence of exposure.

A useful model must distinguish three questions. How many medals are likely, given current national capacity? Which countries benefit if the event portfolio changes? And how much of a performance change can be attributed to coaching? The first two admit conditional calculations from these files. The third has an identification barrier: no coach, hiring date, budget, or training record is observed. We therefore report coach-compatible changes and explicit intervention scenarios, with their assumptions.

1.1 Data audit and identity policy

We inspected the original four-page problem and all five supplied files. The athlete file has 252,565 rows and 1,466 exact duplicates. Medal winners appear once per athlete, sometimes in multiple events. Team suffixes are not interpreted as separate countries: NOC codes are the key. Country names in the medal table are mapped to NOC codes using the athlete file and explicit label aliases. We merge the evidenced Lebanon code rename LIB to LBN; 1992 Independent Olympic Participants maps to IOA, and 1996/2000 FR Yugoslavia to SCG. All positive official rows since 1988 have an athlete-roster match. Beyond evidenced same-country code renames, historical political entities remain separate; there is no automatic transfer of Soviet, East German, Russian, or neutral-team medals to a modern nation.

The 2024 athlete schema differs from older records: it splits several disciplines and includes multi-sport labels. Multi-sport no-medal records contribute exposure to each listed group, with within-group athlete deduplication; no multi-sport medal row is present. We aggregate aquatic disciplines into Aquatics, cycling disciplines into Cycling, and gymnastics disciplines into Gymnastics; historical Equestrianism is mapped to Equestrian; the complete mappings are in the reproducible preparation code. Award units are deduplicated by year, NOC, original sport, event, and medal color. Including sport is essential because generic event names recur in the 2024 records.

For each country and color group, raw sport awards are rescaled to the official country count. This reconciles historical reallocation discrepancies but makes the sport allocation approximate where records disagree. In six country-year/color cells during 2008–2012, an official medal has no matching award of that color. We allocate it across the country's other medal sports, or across participation if no award exists. These six medals are recorded explicitly. The eight unresolved official rows are all before 1988 and do not enter modern forecasts; their presence is retained in the audit.

The supplied 2024 medal table sums to 1,039 medals and 328 golds, and omits Individual Neutral Athletes. Deduplicating the neutral-athlete records adds five medals, including one gold, producing 1,044 and 329. We retain this team separately in the forecast but exclude it and the refugee team from the count of first-medal *nations*. The gold/total forecast is internally coherent even when the source tables are not.

Most importantly, the actual programme file has no 2028 column. It ends in 2024, with 329 events. We cannot infer an official future schedule from its description. The central LA forecast repeats the 2024 programme, while the host file identifies the United States as the 2028 host. Changes are supplied as scenario calculations, not silently introduced future data.

Table 1. Identity decisions illustrated by supplied records.

Original record	Model key	Decision
1992 Independent Olympic Participants	IOA	Edition-specific mapping, not transfer to a nation.
1996/2000 FR Yugoslavia	SCG	Match the observed historical identity.
LIB and LBN	LBN	Evidenced Lebanon code rename; other historical states remain separate.

1.2 Working assumptions

The 2024 roster is the planning roster for 2028; qualification, participation, and political eligibility are conditional assumptions. Recent country-sport performance represents capacity that can persist across an edition, while prior athlete exposure provides a small chance of new success. A host multiplier captures the historical association of hosting with performance, including preparation and selection. Different sports and gold/non-gold allocation shocks are independent in the predictive model. The allocation is an approximation: it does not enforce future qualification or every per-event country entry cap. These assumptions simplify capacity rather than reconstruct individual athlete careers; structural uncertainty is evaluated separately from conditional prediction intervals.

2 A coherent model of country-sport medal opportunities

2.1 Why allocate opportunities rather than regress ranks?

A persistence forecast, $\widehat{T}_{c,t} = T_{c,t-1}$, is a transparent baseline but cannot respond to event changes or assign a first medal to a previous zero. A pooled count regression could use lagged medals and athlete exposure, but would need extra constraints to conserve opportunities and ensure gold does not exceed total. We instead allocate a sport's gold and non-gold pools across countries. This supports a common mechanism for the table, event changes, and investment scenarios.

The same country–sport probabilities enter four calculations: means and complete medal-table draws (Equations 4–5), first-medal absence probabilities (Equation 7), event-pool changes (Equa-

tion 8), and controlled odds scenarios (Equation 10). The last calculation changes an assumed mechanism; it does not estimate a coach effect from treatment data.

Let c index NOCs, s sports, t editions, and $k \in \{G, N\}$ gold and non-gold awards. Write Y^k_{cst} for the reconciled medal count and E_{st} for the number of events. Reconciled counts and forecast means may be fractional; simulated medal outcomes are integer counts. $N^k_{st} = \sum_c Y^k_{cst}$ is the observed pool.

For the next Games, the most recent edition u with sport s supplies its award multiplicity:

$$\widetilde{N}^k_{st} = E_{st} \frac{N^k_{su}}{E_{su}}. \tag{1}$$

Largest-remainder allocation converts these pools to integers, preserving their rounded sum across sports. This accommodates two-bronze sports and historical ties, rather than assuming exactly three medals for every event. A genuinely unseen sport starts with one gold and two non-gold awards per event; because no sport-specific evidence exists, its capacity prior would be much less defensible.

2.2 Recency, exposure, and hosting

The last four editions form a discounted stock of sport capacity:

$$B^k_{cs} = \sum_{j=0}^{3} \delta^j \frac{Y^k_{cs,t-1-j}}{h_{t-1-j}^{\mathbb{1}(c=H_{t-1-j})}}, \qquad 0 < \delta \le 1. \tag{2}$$

H_t is the host and h_t the host strength factor. We use a saturating link, $h_t = 1 + b/(a + T_{H_t,t-1})$, where b and a are in medal units. It gives a roughly additive capacity boost to a large delegation instead of applying the same percentage gain to every host. The prospective host's preceding-edition observed total is used; no target outcome enters the factor. Dividing old host performance by its h_t avoids treating a temporary hosting boost as permanent capacity. Edition indexing, rather than calendar duration, handles the delayed Tokyo Games.

Let q_{cs} be the country's share of distinct athlete-sport participants in the previous edition. For a temporarily absent sport we use its most recent participating edition. If a sport has no observed exposure history, the prior uses the country's overall delegation share. A 0.1% overall-exposure component prevents a historical zero from being a mathematical impossibility. The smoothed allocation probability is

$$p^k_{cs} = \frac{h_t^{\mathbb{1}(c=H_t)} (B^k_{cs} + \lambda_c q_{cs})}{\sum_d h_t^{\mathbb{1}(d=H_t)} (B^k_{ds} + \lambda_d q_{ds})}. \tag{3}$$

The pseudo-count is $\lambda_c = \lambda_Z$ for a nation with no prior official medal and λ otherwise; special teams use λ. Each is shared across sports, not multiplied by an event count. Thus the same evidence has more influence in a small sport. Athlete exposure is measured before the target Games; the model never uses the target athlete roster to predict that edition's performance.

The point forecasts are

$$\mu^G_c = \sum_s N^G_{st} p^G_{cs}, \qquad \mu^T_c = \sum_s \left(N^G_{st} p^G_{cs} + N^N_{st} p^N_{cs} \right). \tag{4}$$

Summing over NOCs gives the entire medal pool exactly. Increasing one nation's probability necessarily reduces others' shares of the same pool.

2.3 Prediction intervals and their interpretation

A plain multinomial would treat national capacity as known and often understate variation. For each active sport–color pool with $N_s^k > 0$, we allow the probability vector to vary. An empty pool has $X_s^k = 0$ deterministically and contributes a factor of one to the no-medal probability; no Dirichlet distribution is drawn for it:

$$P_s^k \sim \text{Dirichlet}(\kappa N_s^k p_s^k), \quad X_s^k \mid P_s^k \sim \text{Multinomial}(N_s^k, P_s^k). \tag{5}$$

This is the Dirichlet-multinomial construction [3]. Its expectation is unchanged, and, with $a_0 = \kappa N_s^k$,

$$\text{Var}(X_{cs}^k) = N_s^k p_{cs}^k (1 - p_{cs}^k) \frac{N_s^k + a_0}{1 + a_0}. \tag{6}$$

We draw gold and non-gold awards and set $T_c = G_c + N_c$. Therefore every draw has $0 \le G_c \le T_c$. The published intervals are marginal 5th and 95th integer quantiles of the predictive distribution; they are not confidence intervals for a mean, nor simultaneous guarantees for every country. Parameter, schedule, qualification, and political uncertainties are outside them.

2.4 Parameter selection and chronological evaluation

The initial baseline selected recency and established-country smoothing on targets 2008, 2012, and 2016, minimizing total MAE plus three times gold MAE. We retained $\delta = 0.35$ and $\lambda = 1$ after correcting all data mappings and repeating that selection. The grids were $\delta \in \{0.35, 0.60, 0.85, 1\}$ and $\lambda \in \{1, 5, 15, 50\}$. This overall error criterion alone strongly favored a conservative prior for previous non-winners, so we calibrated that population separately.

For the host link, six host targets in 1996–2016 compared a percentage multiplier against the saturating link in Equation 2. The host loss was absolute total error plus three times absolute gold error. The selected bonus was $b = 20$ medals, with regularizer $a = 5$. Its mean host loss was 30.57, versus 31.17 for the best constant multiplier. This small improvement supports a planning scenario, not a precisely established host law. Leaving one historical host out changes the selected link: the resulting US 2028 means range from 131.8 to 192.4 medals. The 1996 US experience is particularly influential.

The never-winner prior $\lambda_Z = 10$ was selected from $\{1, 3, 5, 10, 15, 30, 50, 100\}$ using Bernoulli log loss on first-medal outcomes in the same 2008–2016 targets, conditional on each preceding roster. Its mean loss was 0.2270, versus 0.2943 for $\lambda_Z = 1$. This prevents overall MAE from effectively erasing a required rare-event outcome. The predictive dispersion $\kappa = 0.8$, originally selected on development residuals from a moment-based Gaussian score, was held fixed during this sequential calibration. Simulation uses seed 7391858.

The numerical fitting targets precede 2020. Nevertheless, the initial 2020/2024 evaluation feedback was inspected before later structural development and data repairs. These are therefore reused

chronological evaluations, not an untouched final test; the scores can contain model-design selection optimism. For each evaluation, event counts are treated as a known programme, while all performance and exposure inputs precede the target. The scoring universe is the union of forecast and actual NOCs: newly appearing identities receive a zero forecast rather than disappearing from the error measure. The data are a single retrospective vintage and may contain medal reallocations not known at the historical forecast date.

Table 2. Chronological country-level accuracy; D is numerical development and E is reused chronological evaluation.

Year	Gold	Baseline	Total	Baseline	Cov. 90%
2008 (D)	0.64	0.69	1.48	1.60	93.7%
2012 (D)	0.65	0.65	1.25	1.24	97.1%
2016 (D)	0.60	0.59	1.40	1.21	95.7%
2020 (E)	0.90	0.90	2.09	2.19	91.4%
2024 (E)	0.69	0.69	1.48	1.77	94.2%

Table 2 shows that the richer model does not dominate persistence on every development edition. In the two evaluation editions it improved total MAE by 4.4% and 16.5%. The 2024 total RMSE was 4.12 and predictive R^2 was 0.915. The all-country 90% interval coverage was 94.2%, while coverage among actual medallists was 89.1%. Zero-heavy panels can make overall coverage look better than coverage for the countries of greatest interest.

Table 3. Total-medal interval coverage by diagnostic group.

Group	2020 covered/scored	2024 covered/scored
All scored NOCs	192/210 (91.4%)	196/208 (94.2%)
Actual medallists	77/93 (82.8%)	82/92 (89.1%)
Prior edition: zero	112/124 (90.3%)	106/115 (92.2%)
Prior edition: 1–9	53/57 (93.0%)	66/68 (97.1%)
Prior edition: 10+	27/29 (93.1%)	24/25 (96.0%)

Table 3 replays the accepted parameters, seeds, and 5,000 draws per edition without refitting. Groups defined by the preceding edition can be specified before the target; "prior zero" does not mean never a medallist. Actual medallists are an outcome-selected, retrospective diagnostic. Their 2020 coverage of 82.8% exposes a gap behind the 91.4% overall value; it is not a guarantee for a future identifiable set of winners. The scoring roster still includes target-only NOCs after the fact. Width and a central-90% interval score were also retained, with penalties $20(l - y)$ below l and $20(y - u)$ above u. Raw scores vary with medal-count scale, so we do not rank differently sized groups by them. These reused evaluations cannot establish future calibration.

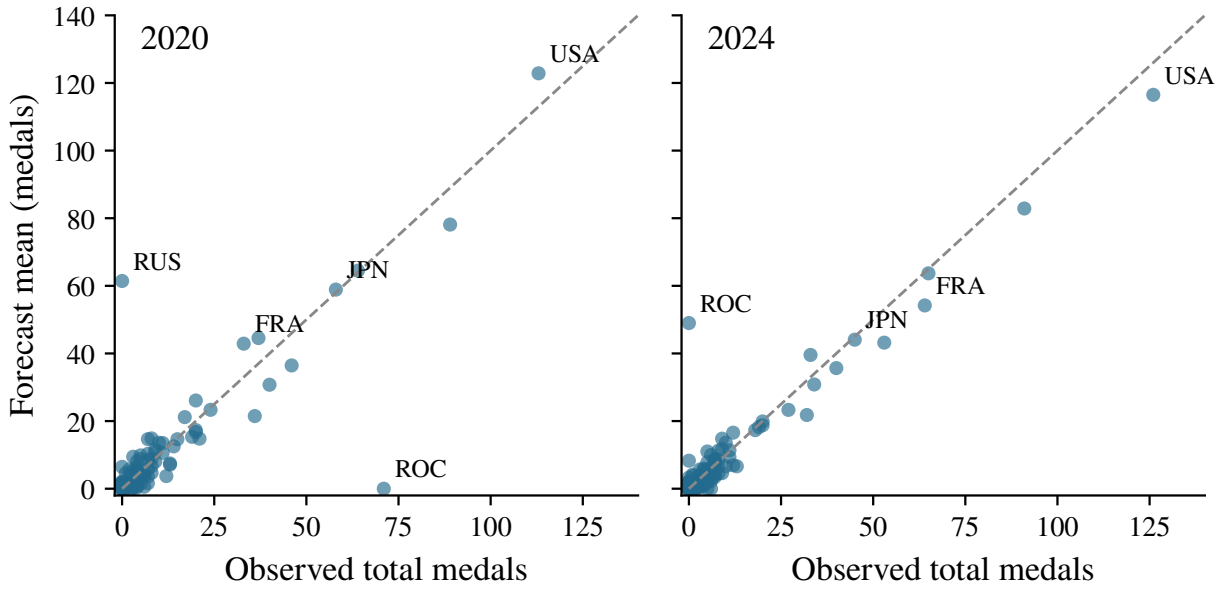


Figure 1. Chronological forecasts preserve the leading-country scale but miss some roster and performance shocks. Each point is a NOC from the union scoring roster; the dashed line is equality.

Figure 1 exposes errors hidden by an average over more than 200 countries. A high historical score for a subsequently absent identity cannot be fixed by a more elaborate medal regression alone. Eligibility scenarios should be maintained explicitly by a committee.

3 Conditional medal table for Los Angeles 2028

3.1 Leading countries and expected changes

Under the unchanged programme, there are 329 gold and 1044 total award opportunities. Table 4 lists the leading fifteen by expected gold, breaking ties by expected total. Decimal means summarize a distribution; they are not fractional actual medals. The complete table for all 206 NOCs appears in Appendix A, including zero-medal countries.

Table 4. 2028 expected medals and marginal 90% prediction intervals; change is expected total minus 2024 total.

Country	Gold [90% PI]	Total [90% PI]	ΔT
United States	45.6 [32, 60]	139.9 [115, 166]	+13.9
China	36.7 [25, 49]	88.1 [69, 109]	-2.9
Japan	19.5 [11, 29]	45.1 [31, 60]	+0.1
Australia	17.0 [9, 27]	49.7 [35, 66]	-3.3
Great Britain	16.9 [9, 26]	64.4 [47, 83]	-0.6
Netherlands	12.6 [6, 21]	32.8 [21, 46]	-1.2
Germany	12.3 [6, 20]	35.8 [23, 50]	+2.8
Italy	11.2 [4, 20]	39.9 [26, 55]	-0.1
France	10.7 [4, 18]	41.8 [28, 57]	-22.2
South Korea	10.6 [5, 17]	28.6 [18, 41]	-3.4
New Zealand	8.5 [3, 15]	19.5 [10, 30]	-0.5
Canada	8.2 [3, 15]	26.0 [15, 38]	-1.0
Hungary	6.3 [2, 13]	19.3 [10, 30]	+0.3
Uzbekistan	6.3 [2, 12]	11.1 [4, 19]	-1.9
Spain	4.7 [1, 10]	18.3 [9, 29]	+0.3

The model favors a strong United States improvement, with probability 80.7% of exceeding 126 medals. France has a 98.9% probability of falling below 64 as its host adjustment expires. These conclusions are strongly conditional on transferring the historical host coefficient to LA. Germany's mean rises by 2.8 medals, but its improvement probability of 58.9% is much less decisive. China, Australia, and South Korea have decline probabilities of 58.8%, 62.4%, and 67.7%; this is probabilistic regression from 2024, not a claim that their athletes will deteriorate.

3.2 The next first-medal nations

A nation is eligible for a first medal if it belongs to the planning roster and has no positive official medal count under its NOC in any supplied edition. The refugee and neutral teams are excluded from this nation count. There are 65 eligible nations. For each country, the zero probability is available without simulation. The beta-binomial marginal of Equation 5 gives

$$\Pr(T_c = 0) = \prod_{s,k} \frac{(a_0 - a_{cs}^k)_{N_s^k}}{(a_0)_{N_s^k}}, \qquad \mathbb{E}[F] = \sum_{c \in \mathcal{Z}} [1 - \Pr(T_c = 0)], \tag{7}$$

where $(x)_n = x(x+1)\cdots(x+n-1)$ and $\mathcal{Z}$ is the eligible set. The sum needs no independence across countries. We obtain $\mathbb{E}[F] = 5.378$, median 5, mode 5, and a 90% prediction interval of [2, 9]. The probability of at least one first medallist is 99.8%. The estimated probability of the modal count 5 is approximately 0.191, as shown in Figure 2.

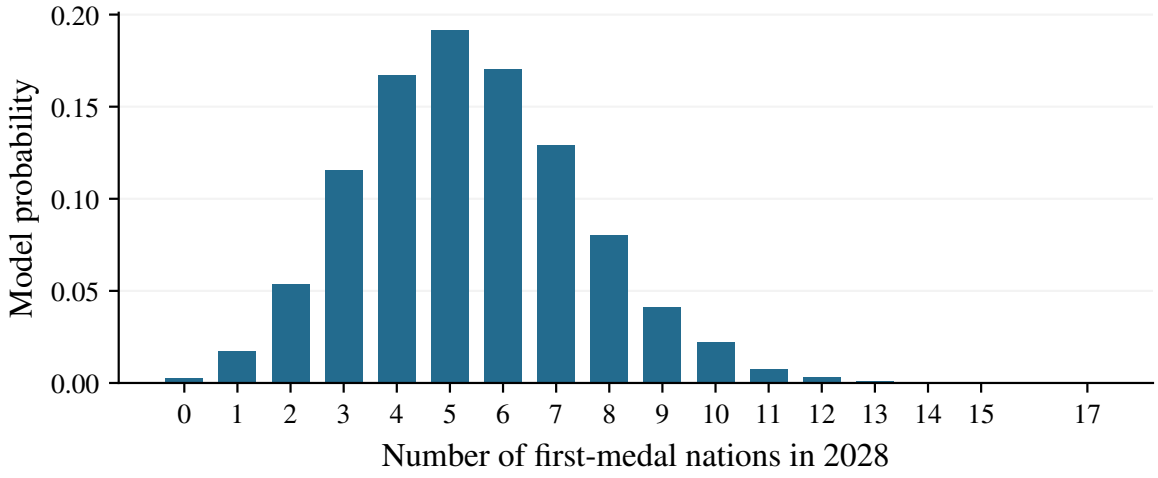


Figure 2. The model assigns a distribution to the number of first-medal nations, rather than rounding the expected value. Probabilities use 24,000 complete, competing country-allocation simulations.

Table 5. Leading first-medal candidates under the fitted exposure prior.

Nation	Pr(any medal)	Expected total	90% PI
Angola	40.1%	0.70	[0, 3]
Guinea	32.0%	0.52	[0, 2]
El Salvador	31.9%	0.53	[0, 2]
Mali	31.0%	0.50	[0, 2]
South Sudan	26.5%	0.43	[0, 2]
Nicaragua	23.4%	0.36	[0, 2]

Table 5 ranks opportunities, not confirmed contenders. Small delegations with previous participation in medal-rich sports receive more prior mass. The first-medal prior is calibrated separately: its development expected counts were about 5.9, 6.2, and 7.3 versus 6, 7, and 2 observed within the prior roster. The last mismatch shows substantial uncertainty. The prior sensitivity should be considered before using it to allocate a national budget; the model does not know emerging individual talent.

4 How the programme changes the medal table

4.1 A country's important sports are a portfolio

For each country, define its expected sport contribution $\mu_{cs}^{T} = N_s^G p_{cs}^G + N_s^N p_{cs}^N$ and its within-country share $w_{cs} = \mu_{cs}^{T}/\mu_c^{T}$. The important sport is the one providing a large medal contribution, while comparative specialization asks whether that sport takes a larger share than it does globally.

These are different questions: a small but specialized sport can be strategically valuable without dominating the total table.

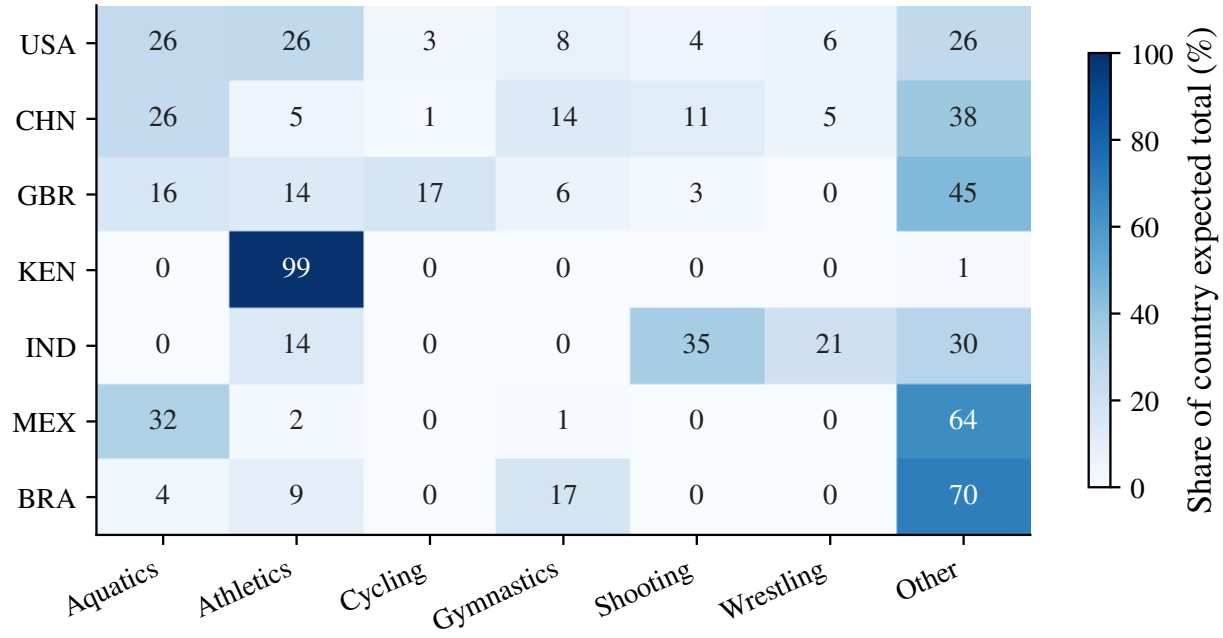


Figure 3. National opportunity portfolios differ sharply. Cells show the percentage of each country's expected total; Other preserves the full row sum. The same number of added events therefore has unequal consequences across countries.

Figure 3 shows why a pooled national growth coefficient is insufficient. Aquatics and athletics support the United States' large total; China's strength is distributed differently; Kenya is concentrated in athletics. Within Aquatics, the original athlete records show that diving supplies Mexico's recent medal contribution. Participation and previous conversion are useful capacity proxies, although they do not establish why one training system is better than another.

Table 6. Two largest expected sport contributions for selected countries.

NOC	Largest sport	Medals	Second sport	Medals
USA	Athletics	36.7	Aquatics	36.2
CHN	Aquatics	22.7	Gymnastics	12.0
GBR	Cycling	11.0	Aquatics	10.4
KEN	Athletics	10.8	Rugby	0.0
IND	Shooting	2.2	Wrestling	1.3
MEX	Aquatics	1.7	Archery	1.0
BRA	Judo	3.5	Gymnastics	3.5

The contributions in Table 6 identify where continuity matters most. They describe expected

country awards, rather than counting the larger number of athletes who stand on a team podium.

4.2 Event number, event type, and a host's choices

Holding estimated capacity fixed, an extra event in sport s changes a country's mean by

$$\frac{\partial \mu_c^T}{\partial E_s} = \rho_s^G p_{cs}^G + \rho_s^N p_{cs}^N, \qquad \rho_s^k = \frac{N_{su}^k}{E_{su}}. \tag{8}$$

This is a marginal opportunity calculation, before integer-pool rounding. It supplies a direct answer to how event choices benefit hosts: choosing events in high-capacity sports increases the host's expected return. At a fixed total event count, replacing a low-return sport with a high-return sport is more valuable to the host than indiscriminate programme expansion. The model does not assume the host has unilateral authority over the programme.

We evaluated four transparent changes drawn from sports in the supplied programme history. Removing the two Breaking events is a subtraction scenario; reinstating baseball/softball uses its two-event archived programme; reinstating karate uses its eight-event archived programme; adding ten athletics events isolates event volume. These are not representations of an official 2028 schedule.

Table 7. Conditional programme changes: added total award pool and United States expected change.

Scenario	All medals	Δ USA medals
Remove Breaking	-6	-1.19
Return baseball/softball	+6	+1.83
Return karate	+32	+1.32
Add ten athletics events	+31	+7.68

Table 7 shows that the event type matters as well as the count. A programme change can distribute many new medals outside the host even if the host has a positive return. For returning sports, history is older and roster exposure is stale; the scenario should guide what evidence to collect rather than lock in a budget. Every added country medal in a fixed-pool coaching scenario comes at another country's expense, whereas programme expansion creates additional opportunities.

5 What can be learned about a great-coach effect?

5.1 A change screen with a clear identification limit

A coach may affect training and tactics, but medal records also respond to athlete cohorts, participation, financing, selection, rule changes, and reallocated medals. Write

$$R_{cst} = \beta C_{cst} + A_{cst} + I_{cst} + \epsilon_{cst}, \tag{9}$$

where R is performance, C coach quality, A athlete capacity, and I investment and institutions. Only R and partial exposure are observed. For any value of βC, the same observation can be reproduced by a compensating $A + I$. Coach contribution is therefore not point-identifiable from these files. The Lang Ping and Károlyi examples in the problem motivate the mechanism but do not add a coach-treatment variable to the supplied data.

We instead screen for changes *compatible* with sustained improvement. For split years 2000, 2004, 2008, 2012, and 2016, we compare two editions before and two after. Divide country-sport medals by event count, adjust the pre-period level by the post/pre athlete-exposure ratio (with two-participant smoothing and ratio clipped to [0.5,2]), then express the excess in the final post-period programme's medal units. We require at least three athlete-sport participants per period, both post editions above the pre-period rate, and no host for the country in the four-edition window. This exploratory screen controls some obvious alternatives but is not a causal design or a hypothesis test after selection.

Table 8. Largest exploratory sustained gains compatible with coaching or other institutional improvements.

NOC	Sport	Split	Pre	Post	Residual gain
NED	Cycling	2016	2.44	9.67	7.04
NED	Aquatics	2000	1.16	7.50	6.23
KEN	Athletics	2008	7.15	14.50	6.14
JPN	Aquatics	2000	2.32	8.00	5.93
RUS	Gymnastics	2012	5.28	10.27	5.70
GBR	Cycling	2004	3.29	9.31	5.59

Table 9. Women's indoor volleyball and artistic gymnastics in the supplied athlete records; entries are gold/total country award units.

Year	USA volley.	CHN volley.	USA gymnast.	ROU gymnast.
2000	0/0	0/0	0/1	2/5
2004	0/0	1/1	1/6	4/6
2008	0/1	0/1	2/8	1/2
2012	0/1	0/0	3/5	1/3
2016	0/1	1/1	4/9	0/0
2020	1/1	0/0	2/6	0/0
2024	0/1	0/0	3/7	0/1

Table 9 examines the specific sports raised by the problem. The US women's indoor volleyball team earned a medal in every listed edition from 2008 onward, while China returned to gold in

2016. US women's artistic gymnastics rose to nine award units in 2016 as Romania fell to zero. Such reversals are compatible with coaching or institutional changes, but the records do not identify who coached each squad. We do not assign a causal share of these medals to Lang Ping or Károlyi without a treatment record. These raw award-unit examples are separate from the reconciled national forecasting targets.

Table 8 provides candidate investigations rather than named successful coaches. Netherlands cycling and Great Britain cycling exhibit substantial sustained improvements; the appropriate next step is to obtain treatment dates and matched athlete-development evidence. Multiple overlapping screens and the absence of treatment labels prevent interpreting the maximum residual as an unbiased coach estimate. Even a positive residual could be entirely athlete-driven; a causal contribution of zero remains compatible with these data.

5.2 Three investment proposals and their conditional impact

We select India shooting, Mexico diving, and Brazil gymnastics for investigation because the supplied records show recent participation and medal conversion. Those observable proxies do not establish training infrastructure, funding, or coach quality. A coaching proposal should specify the technical improvement to be measured, rather than treat a larger delegation as an intervention effect. For a sport probability p and odds multiplier m, a controlled scenario is

$$p' = \frac{mp}{1 + (m-1)p}, \qquad \Delta\mu_c^T = N_s^G(p'_G - p_G) + N_s^N(p'_N - p_N). \tag{10}$$

The competing countries are renormalized, so the scenario conserves medals. The multiplier is a stated intervention assumption, not an estimated causal coefficient. For Mexico, all recent Aquatics medal units in the supplied 2016–2024 records were in diving. The computed scenario nevertheless raises odds for the aggregated Aquatics group, including other disciplines; its gain is therefore an Aquatics proxy, not an identified diving-only effect. A diving intervention requires a finer model and corresponding exposure evidence.

Table 10. Coach-investment scenarios. Central gains assume a 25% increase in sport odds; the range uses 10–50%, not a confidence interval.

Country	Sport	Baseline	Gain total	Gain gold	Gain range
India	Shooting	2.19	0.50	0.01	0.20–0.99
Mexico	Aquatics	1.69	0.41	0.00	0.17–0.82
Brazil	Gymnastics	3.51	0.81	0.23	0.33–1.60

Table 10 quantifies modest expected gains: an improvement of half a medal means a change in probabilities, not a guaranteed podium. India can investigate shooting conversion, Mexico can investigate diving through the stated coarse proxy, and Brazil can assess gymnastics coaching and

succession. These are candidates for committee scrutiny, not measured diagnoses of their training systems. Country-specific gold gains may be small because the separate gold stock contains limited recent gold evidence. Committees should compare a coach's full cost with the value of the estimated probability gain, and measure intermediate technical outcomes before attributing a later medal to the hire. The supplied data contain no budget or coach salaries, so no monetary return on investment is claimed.

6 Original committee insights: protect opportunities, not just ranks

A medal-table position compresses a portfolio into one ordinal number. We use $\mathrm{HHI}_c = \sum_s w_{cs}^2$ and $1/\mathrm{HHI}_c$ as the effective number of sports supporting a country's expected medals. A country can have a reasonable expected total yet depend heavily on one event group. Its vulnerability to a programme cut is measurable separately from its current rank.

Table 11. Portfolio concentration and a 20% cut to the largest sport event opportunity, with capacity held fixed.

NOC	Effective sports	Largest sport	Share	Expected loss
USA	6.5	Athletics	26.3%	7.35
CHN	8.5	Aquatics	25.8%	4.54
KEN	1.0	Athletics	99.1%	2.15
IND	4.7	Shooting	35.1%	0.44
MEX	5.3	Aquatics	32.3%	0.34
BRA	9.4	Judo	17.4%	0.71

Table 11 suggests two different actions. Countries with broad portfolios should prioritize continuity and several high-return opportunities; countries concentrated in one sport should insure their technical pipeline and develop one feasible second pillar. Diversification is not automatically optimal: a weak new sport may be more costly than protecting a strong existing programme. The quantitative cut calculation gives a risk measure, not a prescription to spend equally across sports.

A second insight is that marginal event advantage and total medal strength are distinct. A host can lead the total table while gaining little from an added sport where its capacity share is weak. Conversely, a smaller delegation can receive a disproportionately large benefit from a specific returning sport. Olympic committees should maintain a sport-by-sport opportunity ledger, with a forecast conditional on each plausible event programme, rather than issue a single unqualified medal target.

7 Verification, sensitivity, and limits

7.1 Independent numerical evidence

The stored validation suite contains 21 checks, of which 21 passed. It checks coherent integer draws, exact medal-pool conservation, probability normalization, official first-medal history, chronological target separation, roster policy, interval ordering, a raw team-medal example, and a separate reduction of the backtest MAE. Independent finite-product calculations of zero-medal probabilities were compared with simulated absence rates; theoretical Dirichlet-multinomial means and variances were compared with the actual 24,000 draws. These are numerical verification and chronological predictive evidence, not certification of future reality.

7.2 Structural sensitivity and decisions that can change

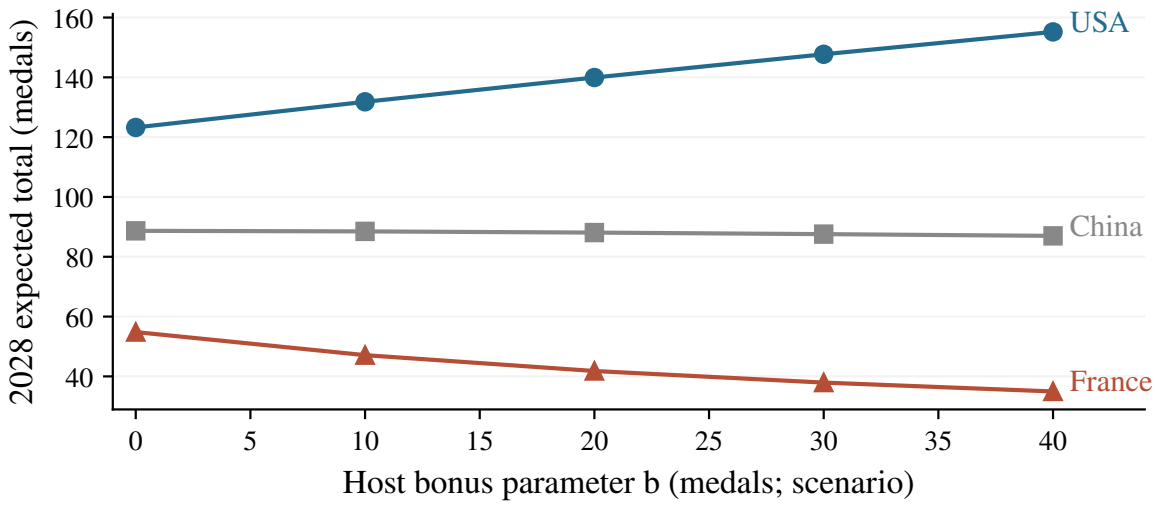


Figure 4. The host-transfer assumption materially changes the LA projection. Curves are deterministic scenario means with other fitted parameters fixed; they are not causal response estimates or statistical confidence bands.

Figure 4 shows why the United States increase must be treated conditionally. The same saturating host link both removes the French boost from the 2024 capacity stock and adds a US boost in 2028. This amplifies the relative shift. The first-medal expected count is also sensitive to its separately calibrated pseudo-count: increasing the prior assigns more mass to small delegations. The supplementary machine-readable results retain all tested decay, pseudo-count, and host scenarios rather than showing only favorable values.

The main model's scope is a useful allocation forecast. It does not forecast athlete injuries, individual qualification, new national training programmes, or returning political entities. Its event categories aggregate unlike disciplines and its within-sport allocation does not enforce all event-level entry limits, and six historical missing-color cells require probabilistic attribution. Gold and

non-gold shocks are independent, so the common capacity shock across colors and sports is incomplete. Future roster and programme changes can dominate some country intervals. The coach analysis cannot isolate a causal contribution; the hiring gains in Table 10 are scenario outputs.

8 Conclusion

A coherent country-sport allocation provides gold and total projections, intervals, first-medal probabilities, and programme sensitivities with a common set of assumptions. The reused chronological evaluations improve over country persistence on 2020 and 2024 totals, although some development editions favor the simpler baseline. The United States leads the conditional 2028 table and France loses its temporary host boost. The appropriate committee response is to update the roster and event programme before treating the numerical table as a commitment, protect the sports carrying its medal portfolio, and evaluate coaching hires through measurable conversion gains rather than retrospective attribution alone.

References

[1] COMAP. *2025 MCM Problem C: Models for Olympic Medal Tables*. Supplied problem statement, four pages, 2025.

[2] COMAP. *2025 Problem C Data*. Supplied archive containing data dictionary, summer Olympic athletes, medal counts, hosts, and programmes. Numerical data accessed locally on 9 October 2026. The dictionary reports its programme source as Wikipedia; we used only the supplied files.

[3] The SciPy community. *scipy.stats.dirichlet_multinomial*. SciPy reference documentation, accessed 9 October 2026. SciPy distribution reference. Used to check the compound-distribution definition, not as Olympic data.

[4] OpenAI. *Codex AI assistant*. Autonomous model development, Python implementation, verification, figures, and English report generation, 9 October 2026. Exact serving model identifier was not exposed to this executor. See the appended Report on Use of AI Tools.

[5] Team 2510862. *2028 Olympic Medal Predictions: Multi-Effect Analysis and Strategic Insights*. 2025 MCM Problem C paper. Locally obtained watermarked copy, 25 pages; original submission bytes unconfirmed. Post-seal presentation comparison only.

[6] Team 2503389. *Olympic Medal Prediction Model Based on Regression and Machine Learning*. 2025 MCM Problem C paper. Locally obtained watermarked copy, 26 pages; original submission bytes unconfirmed. Post-seal presentation comparison only.

A Complete conditional 2028 forecast

Every planning-roster NOC is included. Entries are mean [5th, 95th predictive percentile]; all intervals are marginal 90% intervals. NOCs follow the supplied athlete file, including the refugee

(EOR) and neutral (AIN) teams. Full country names, change probabilities, and sport breakdowns are retained in the accompanying reproducible CSV. The NOC codes identify countries unambiguously within the supplied data. The table uses expected gold ordering only in the main text; this appendix is alphabetic for lookup.

NOC	Gold [90%]	Total [90%]	NOC	Gold [90%]	Total [90%]
AFG	0.0 [0, 0]	0.0 [0, 0]	AIN	0.8 [0, 3]	3.5 [0, 9]
ALB	0.0 [0, 0]	1.4 [0, 5]	ALG	1.4 [0, 5]	2.3 [0, 6]
AND	0.0 [0, 0]	0.0 [0, 0]	ANG	0.3 [0, 2]	0.7 [0, 3]
ANT	0.0 [0, 0]	0.1 [0, 0]	ARG	1.1 [0, 4]	3.4 [0, 8]
ARM	0.1 [0, 1]	4.0 [0, 9]	ARU	0.1 [0, 1]	0.2 [0, 1]
ASA	0.0 [0, 0]	0.0 [0, 0]	AUS	17.0 [9, 27]	49.7 [35, 66]
AUT	1.6 [0, 5]	5.3 [1, 11]	AZE	1.6 [0, 5]	7.9 [2, 15]
BAH	0.6 [0, 3]	0.6 [0, 3]	BAN	0.1 [0, 0]	0.2 [0, 1]
BAR	0.0 [0, 0]	0.0 [0, 0]	BDI	0.0 [0, 0]	0.1 [0, 1]
BEL	3.0 [0, 8]	9.3 [3, 17]	BEN	0.0 [0, 0]	0.1 [0, 0]
BER	0.2 [0, 1]	0.2 [0, 1]	BHU	0.1 [0, 0]	0.1 [0, 1]
BIH	0.0 [0, 0]	0.1 [0, 1]	BIZ	0.0 [0, 0]	0.0 [0, 0]
BOL	0.0 [0, 0]	0.0 [0, 0]	BOT	0.6 [0, 3]	1.6 [0, 5]
BRA	4.3 [1, 9]	20.4 [11, 31]	BRN	1.4 [0, 5]	3.0 [0, 8]
BRU	0.0 [0, 0]	0.0 [0, 0]	BUL	2.5 [0, 7]	6.2 [1, 13]
BUR	0.0 [0, 0]	0.3 [0, 2]	CAF	0.0 [0, 0]	0.1 [0, 0]
CAM	0.0 [0, 0]	0.0 [0, 0]	CAN	8.2 [3, 15]	26.0 [15, 38]
CAY	0.0 [0, 0]	0.1 [0, 0]	CGO	0.0 [0, 0]	0.1 [0, 1]
CHA	0.1 [0, 0]	0.1 [0, 1]	CHI	0.7 [0, 3]	1.5 [0, 5]
CHN	36.7 [25, 49]	88.1 [69, 109]	CIV	0.1 [0, 1]	1.1 [0, 4]
CMR	0.0 [0, 0]	0.0 [0, 0]	COD	0.1 [0, 1]	0.2 [0, 1]
COK	0.0 [0, 0]	0.0 [0, 0]	COL	0.4 [0, 2]	4.9 [1, 11]
COM	0.0 [0, 0]	0.1 [0, 0]	CPV	0.0 [0, 0]	0.7 [0, 3]
CRC	0.0 [0, 0]	0.0 [0, 0]	CRO	2.6 [0, 7]	7.7 [2, 15]
CUB	3.6 [0, 8]	11.0 [4, 20]	CYP	0.0 [0, 0]	0.7 [0, 3]
CZE	3.2 [0, 8]	7.4 [2, 14]	DEN	2.1 [0, 5]	10.0 [4, 18]
DJI	0.0 [0, 0]	0.0 [0, 0]	DMA	0.6 [0, 3]	0.6 [0, 3]
DOM	0.7 [0, 3]	3.2 [0, 8]	ECU	1.1 [0, 4]	3.9 [0, 9]
EGY	0.8 [0, 3]	3.7 [0, 9]	EOR	0.1 [0, 1]	0.9 [0, 4]
ERI	0.0 [0, 0]	0.0 [0, 0]	ESA	0.2 [0, 2]	0.5 [0, 2]

NOC	Gold [90%]	Total [90%]	NOC	Gold [90%]	Total [90%]
ESP	4.7 [1, 10]	18.3 [9, 29]	EST	0.3 [0, 2]	0.7 [0, 3]
ETH	1.0 [0, 4]	4.4 [0, 10]	FIJ	0.3 [0, 2]	1.2 [0, 4]
FIN	0.1 [0, 1]	0.8 [0, 3]	FRA	10.7 [4, 18]	41.8 [28, 57]
FSM	0.0 [0, 0]	0.0 [0, 0]	GAB	0.0 [0, 0]	0.0 [0, 0]
GAM	0.1 [0, 1]	0.2 [0, 1]	GBR	16.9 [9, 26]	64.4 [47, 83]
GBS	0.1 [0, 1]	0.2 [0, 2]	GEO	2.5 [0, 7]	7.3 [2, 14]
GEQ	0.0 [0, 0]	0.0 [0, 0]	GER	12.3 [6, 20]	35.8 [23, 50]
GHA	0.0 [0, 0]	0.2 [0, 2]	GRE	1.4 [0, 5]	7.0 [2, 14]
GRN	0.0 [0, 0]	1.6 [0, 5]	GUA	0.7 [0, 3]	1.4 [0, 5]
GUI	0.2 [0, 2]	0.5 [0, 2]	GUM	0.2 [0, 1]	0.3 [0, 2]
GUY	0.0 [0, 0]	0.0 [0, 0]	HAI	0.0 [0, 0]	0.0 [0, 0]
HKG	1.7 [0, 5]	4.0 [0, 9]	HON	0.0 [0, 0]	0.1 [0, 0]
HUN	6.3 [2, 13]	19.3 [10, 30]	INA	1.6 [0, 5]	3.4 [0, 8]
IND	0.4 [0, 2]	6.3 [1, 13]	IRI	3.0 [0, 7]	10.6 [4, 19]
IRL	3.3 [0, 8]	6.2 [1, 13]	IRQ	0.0 [0, 0]	0.0 [0, 0]
ISL	0.0 [0, 0]	0.0 [0, 0]	ISR	1.2 [0, 4]	6.1 [1, 13]
ISV	0.0 [0, 0]	0.0 [0, 0]	ITA	11.2 [4, 20]	39.9 [26, 55]
IVB	0.0 [0, 0]	0.1 [0, 0]	JAM	2.1 [0, 6]	7.2 [2, 14]
JOR	0.1 [0, 1]	1.1 [0, 4]	JPN	19.5 [11, 29]	45.1 [31, 60]
KAZ	1.1 [0, 4]	8.2 [2, 16]	KEN	4.0 [0, 10]	10.9 [4, 19]
KGZ	0.0 [0, 0]	4.8 [1, 11]	KIR	0.1 [0, 0]	0.1 [0, 1]
KOR	10.6 [5, 17]	28.6 [18, 41]	KOS	0.6 [0, 3]	2.0 [0, 6]
KSA	0.0 [0, 0]	0.1 [0, 0]	KUW	0.0 [0, 0]	0.3 [0, 2]
LAO	0.0 [0, 0]	0.1 [0, 0]	LAT	0.2 [0, 2]	0.5 [0, 3]
LBA	0.1 [0, 1]	0.2 [0, 1]	LBN	0.0 [0, 0]	0.0 [0, 0]
LBR	0.0 [0, 0]	0.1 [0, 0]	LCA	0.6 [0, 3]	1.3 [0, 5]
LES	0.1 [0, 0]	0.1 [0, 1]	LIE	0.0 [0, 0]	0.0 [0, 0]
LTU	0.1 [0, 1]	3.7 [0, 9]	LUX	0.0 [0, 0]	0.1 [0, 0]
MAD	0.1 [0, 1]	0.2 [0, 2]	MAR	1.0 [0, 4]	1.8 [0, 5]
MAS	0.1 [0, 0]	2.3 [0, 6]	MAW	0.0 [0, 0]	0.0 [0, 0]
MDA	0.1 [0, 0]	3.1 [0, 8]	MDV	0.1 [0, 1]	0.2 [0, 1]
MEX	0.2 [0, 2]	5.2 [1, 11]	MGL	0.1 [0, 0]	2.1 [0, 6]
MHL	0.1 [0, 0]	0.1 [0, 1]	MKD	0.0 [0, 0]	0.3 [0, 2]
MLI	0.2 [0, 2]	0.5 [0, 2]	MLT	0.0 [0, 0]	0.1 [0, 1]
MNE	0.0 [0, 0]	0.1 [0, 0]	MON	0.1 [0, 1]	0.2 [0, 1]
MOZ	0.0 [0, 0]	0.0 [0, 0]	MRI	0.0 [0, 0]	0.0 [0, 0]

NOC	Gold [90%]	Total [90%]	NOC	Gold [90%]	Total [90%]
MTN	0.0 [0, 0]	0.0 [0, 0]	MYA	0.0 [0, 0]	0.1 [0, 1]
NAM	0.0 [0, 0]	0.2 [0, 2]	NCA	0.2 [0, 1]	0.4 [0, 2]
NED	12.6 [6, 21]	32.8 [21, 46]	NEP	0.1 [0, 1]	0.2 [0, 2]
NGR	0.1 [0, 1]	0.7 [0, 3]	NIG	0.0 [0, 0]	0.1 [0, 1]
NOR	3.3 [0, 8]	7.3 [2, 14]	NRU	0.0 [0, 0]	0.0 [0, 0]
NZL	8.5 [3, 15]	19.5 [10, 30]	OMA	0.0 [0, 0]	0.1 [0, 0]
PAK	0.6 [0, 3]	0.7 [0, 3]	PAN	0.0 [0, 0]	0.7 [0, 3]
PAR	0.0 [0, 0]	0.1 [0, 0]	PER	0.1 [0, 0]	0.8 [0, 3]
PHI	1.6 [0, 5]	3.8 [0, 9]	PLE	0.1 [0, 1]	0.3 [0, 2]
PLW	0.0 [0, 0]	0.0 [0, 0]	PNG	0.2 [0, 1]	0.4 [0, 2]
POL	2.2 [0, 6]	11.9 [5, 21]	POR	1.0 [0, 4]	4.1 [0, 10]
PRK	0.3 [0, 2]	4.7 [1, 11]	PUR	0.4 [0, 2]	1.9 [0, 6]
QAT	0.4 [0, 2]	1.4 [0, 5]	ROU	2.5 [0, 7]	7.6 [2, 15]
RSA	1.4 [0, 5]	6.0 [1, 12]	RWA	0.1 [0, 0]	0.1 [0, 1]
SAM	0.0 [0, 0]	0.1 [0, 1]	SEN	0.0 [0, 0]	0.0 [0, 0]
SEY	0.0 [0, 0]	0.0 [0, 0]	SGP	0.1 [0, 1]	0.9 [0, 4]
SKN	0.0 [0, 0]	0.0 [0, 0]	SLE	0.0 [0, 0]	0.1 [0, 0]
SLO	2.3 [0, 6]	4.0 [0, 9]	SMR	0.0 [0, 0]	0.7 [0, 3]
SOL	0.0 [0, 0]	0.0 [0, 0]	SOM	0.0 [0, 0]	0.0 [0, 0]
SRB	2.7 [0, 7]	6.3 [1, 13]	SRI	0.0 [0, 0]	0.0 [0, 0]
SSD	0.2 [0, 1]	0.4 [0, 2]	STP	0.0 [0, 0]	0.1 [0, 1]
SUD	0.0 [0, 0]	0.0 [0, 0]	SUI	1.8 [0, 5]	9.2 [3, 17]
SUR	0.0 [0, 0]	0.0 [0, 0]	SVK	0.4 [0, 2]	2.1 [0, 6]
SWE	3.6 [0, 9]	10.7 [4, 19]	SWZ	0.0 [0, 0]	0.0 [0, 0]
SYR	0.0 [0, 0]	0.2 [0, 2]	TAN	0.0 [0, 0]	0.0 [0, 0]
TGA	0.0 [0, 0]	0.0 [0, 0]	THA	1.1 [0, 4]	5.0 [1, 11]
TJK	0.1 [0, 1]	2.2 [0, 7]	TKM	0.0 [0, 0]	0.2 [0, 2]
TLS	0.1 [0, 0]	0.1 [0, 1]	TOG	0.0 [0, 0]	0.0 [0, 0]
TPE	1.9 [0, 5]	7.6 [2, 15]	TTO	0.0 [0, 0]	0.2 [0, 1]
TUN	1.0 [0, 4]	3.0 [0, 8]	TUR	0.7 [0, 3]	8.6 [3, 16]
TUV	0.0 [0, 0]	0.0 [0, 0]	UAE	0.0 [0, 0]	0.1 [0, 1]
UGA	1.1 [0, 4]	2.2 [0, 7]	UKR	2.8 [0, 7]	14.2 [6, 24]
URU	0.0 [0, 0]	0.1 [0, 1]	USA	45.6 [32, 60]	139.9 [115, 166]
UZB	6.3 [2, 12]	11.1 [4, 19]	VAN	0.1 [0, 1]	0.2 [0, 2]
VEN	0.3 [0, 2]	1.3 [0, 5]	VIE	0.1 [0, 1]	0.3 [0, 2]
VIN	0.0 [0, 0]	0.0 [0, 0]	YEM	0.0 [0, 0]	0.1 [0, 1]
ZAM	0.0 [0, 0]	0.7 [0, 3]	ZIM	0.0 [0, 0]	0.0 [0, 0]

Report on Use of AI Tools

OpenAI Codex was used on 9 October 2026 throughout the work: interpreting the supplied problem, auditing and reconciling files, choosing the mathematical representation, writing Python, checking numerical results, drawing figures, interpreting limitations, and drafting the English report [4]. The work was an autonomous AI study under human-defined scope and budget. No human mathematical verification is claimed. The serving model identifier and exact token use were unavailable.

The operative instruction was to complete the original 2025 MCM C research using only the supplied inputs and frozen Praxis methods, avoid same-problem answers, preserve the first complete draft before improvement, and produce a reproducible English paper with actual PDF review. The assistant generated the mathematical model, code, computed result files, and this report. The outputs are represented by those saved artifacts, not by a claim that AI merely polished language.

The assistant inspected the supplied original problem, data dictionary, all provided data files, and operational references from the frozen Praxis plugin. During the sealed autonomous stage it did not consult same-problem papers, purchased solutions, previous attempts, or judges' comments. One SciPy reference page was used to verify the Dirichlet-multinomial definition. The COMAP AI policy linked from the original problem currently resolves to a document marked v102025; its present version was read for disclosure context, but was not represented as a verified January 2025 policy copy. No external Olympic numerical data entered the models.

The available record includes scripts, run commands, timestamps, source/input/output hashes, development grids, failures, numerical checks, the initial draft, and final artifacts. It is not a verbatim complete AI conversation transcript. We do not manufacture missing queries or outputs. The modelling and interpretation were checked by the AI executor through explicit independent calculations and reused chronological data. An internal parent AI reviewer independently compared the initial source with the original CSVs, identified three unmapped positive official records, and clarified the evaluation boundary. The executor repaired the mappings and recomputed the affected chain, with their limits stated in the solution. Formal historical AI-policy compliance beyond the supplied problem's explicit format requirements remains unverified.

After the autonomous version was sealed, this separate learning revision critically compared locally obtained papers from teams 2510862 and 2503389 [5, 6]. It uses their result-navigation and data-explanation ideas, not their data, figures, coefficients, or causal claims. The executor added a retrospective subgroup check using the already accepted parameters, removed an unsupported institutional claim, and clarified the Mexico proxy. A distinct AI reviewer independently reduced the new CSV and checked its mathematical interpretations; this is not human verification or an unseen evaluation. The original sealed source and forecasts remain unchanged.